Dragon8 + HVLP Application Note

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1. Introduction

This document describes the processes for identifying the source of a torque limit in genus products. ‘Genus’ is a term used to describe the software present in HVLP and Dragon8. These products share some generic code and functionality, for example TracApp and NVM - as well as some specific modules only applicable to themselves.

Sometimes a system may not provide as much torque output as expected and this can be for a variety of reasons from motor control torque limits, thermals, power limits, speed limits and various other cutbacks.

Genus products have a torque cutback flag that can indicate where the limit is being applied. A torque cutback flag of 0 indicates there are no cutbacks. The rest of the document describes the various flags that can be encountered and how to reduce the effects of the limits to achieve the expected torque output.



A NOTE indicates important information that helps you make better use of your computer system.



A CAUTION indicates either potential damage to hardware or loss of data and tells you how to avoid the problem.



A WARNING indicates the potential for bodily harm and tells you how to avoid the problem.

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2. Motor Control Torque Limits

2.1. Limit Indication

If the available torque is being reduced from the maximum specified in Peak Torque (0x6076, 0) then a torque cutback flag is set.

Peak Torque is multiplied by the motor temperature thermal cutback factor to create a torque limit. The Peak Torque can be viewed and modified in DVT Helper in Tree → Motor → Peak Torque with the units in Newton Meters (Nm). Additionally, the Torque Cutback flag is map-able as a TPDO in 0x4602, sub-index 39 to be viewed in the Vehicle Interface as seen in (Figure 1 Vehicle Interface with Torque Cutback Flag configured as a TPDO) and in the H-protocol as Limit Code in message 11900.

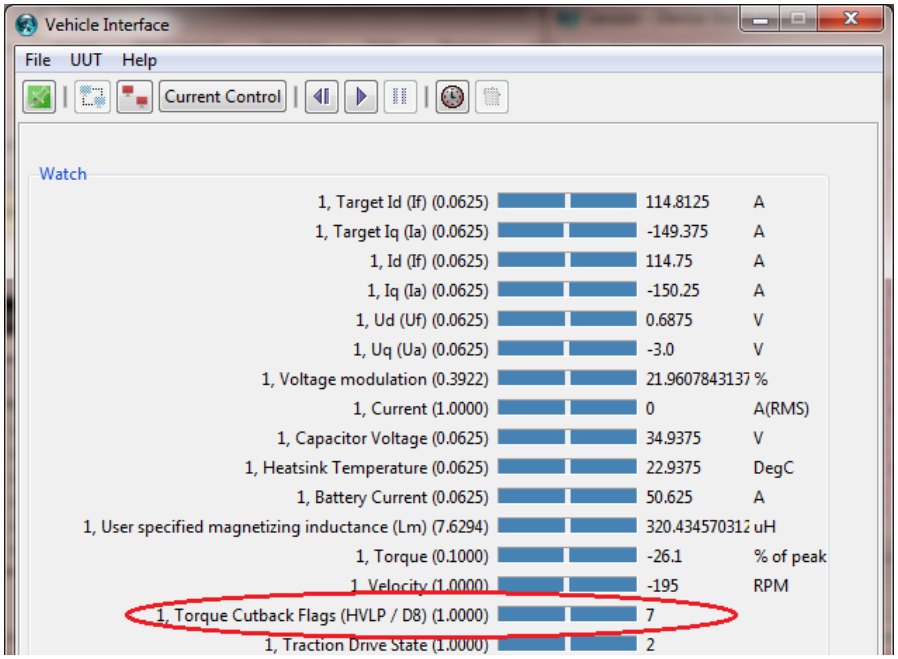


Figure 1 Vehicle Interface with Torque Cutback Flag configured as a TPDO

The below table represents the possible torque cutback flags that can be set by the controller. Zero means that no limit is in effect but the other main possible reasons could be power limits, thermals, user-defined limits, battery voltage cutbacks and speed limits. 3D Lookup table limits only apply to motors that use this as a motor control scheme.

Value	Meaning
0	No limit is in effect
1	Max power limit
2	Motor power limit map
3	Thermals
4	User limits received via H-Protocol
5	3D lookup table calculated limit (specific only to 3D lookup table control schemes)
6	Bridge is not enabled
7	DC (battery) voltage cutback map
8	DC (battery) current cutback
9	Speed limit

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2.2.Local Limits

It is worth checking that the local limits are set to appropriate values. This can be done using the DVT Helper in which Main → Local Limits will display the following; or you can configure the objects via SDO:

Item	Object	Unit	Meaning
Maximum torque ramp up rate	0x6087, 1	Nm/s	Maximum rate of change of torque. Torque ramping is handled by the Master, so this is only a safety limit. Object resolution is Nm/sec
Maximum torque ramp down rate	0x6087, 2	Nm/s	Maximum rate of change for decreasing torque.
Peak Torque	0x6076, 0	Nm	Rated motor torque in mNm.
Maximum motor speed	0x6080, 0	RPM	Maximum motor speed (ωnom) in RPM. It is unsigned and applies in both directions.
Maximum acceleration rate	0x6083, 1	RPM/s	Maximum acceleration and deceleration rates. Acceleration and Deceleration is controlled by the Master, so this is only a safety limit. Object format is 32.0 fixed point, units are rpm/s.
Maximum deceleration rate	0x6083, 2	RPM/s	Maximum acceleration and deceleration rates. Acceleration and Deceleration is controlled by the Master, so this is only a safety limit. Object format is16.0 fixed point, units are rpm/s.
Maximum change in acceleration rate	0x6083, 3	RPM/s	This parameter limits the rate of change of acceleration rate for increasing magnitude of speed demand. It is used to create an s-curve profile for the acceleration rate. (induction V/f only)

The flag applies to the torque direction being demanded, or if zero torque is demanded then the torque direction is being applied by the speed loop. It is important to note that this only applies to the limits originating from the motor control blocks and not does include any limits or cutbacks that TracApp or a VCU calculates for itself; these are discussed later in the document.

If you have a value of zero, but are experiencing a limit, check the following:

- Check that a demand is actually being received by checking that CAN control messages are being received at the correct rate (i.e. not timing out), CAN message checksums and counter values are correct, and the control-word is in the correct state.
- Check the H-Protocol status flags for more information, check the H-Protocol message timeout parameter, and disable checksum and counter checks if necessary.
- Check that peak torque is not set to zero at object 0x6076.
- If torque demand is zero, there isn't a limit. Check that TracApp or a VCU is actually producing a torque demand for the motor control to act on.

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3. Flag 1 - Max power limit

The maximum power limit is fixed for each product variant. It is calculated in terms of mechanical power i.e. a maximum torque for a given speed. The limit does not mean that the product is able to deliver this much power to a motor. In most cases the limit has just been set so that it never acts to reduce the available torque as this limit isn't required for controller protection:

- 60kW Dragon8 72V
- 60kW Dragon8 96V
- 60kW HVLP air cooled
- 60kW HVLP water cooled
- 200kW HDSRDS

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4. Flag 2- Motor power limit map

Torque may be limited by the power limit map (object 0x4611) which relates torque (Nm) to speed (RPM). This is a single configurable motor power limit map settable in the DVT helper Figure 2 Torque maps in DVT helper) and should be set according to the motor capabilities. When selecting torque maps, the Genus power limit map is 0x4611; the other torque maps (2 and 3) apply to other controller types.

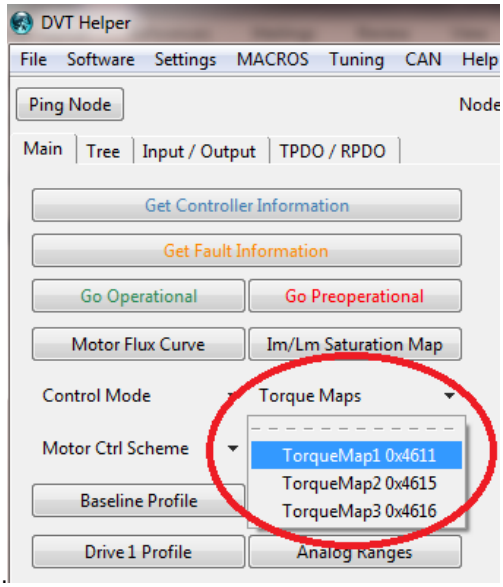


Figure 2 Torque maps in DVT helper

If you find that the power limit map is responsible for torque cutbacks, the map can be configured in the torque map configuration Figure 3 Motor power limit map in DVT Helper). The power limit map has a set of 9 speed and torque limits and these can be changed in the input boxes next to the values. When changing the values it is important to keep the values in ascending order to ensure a valid power limit map.

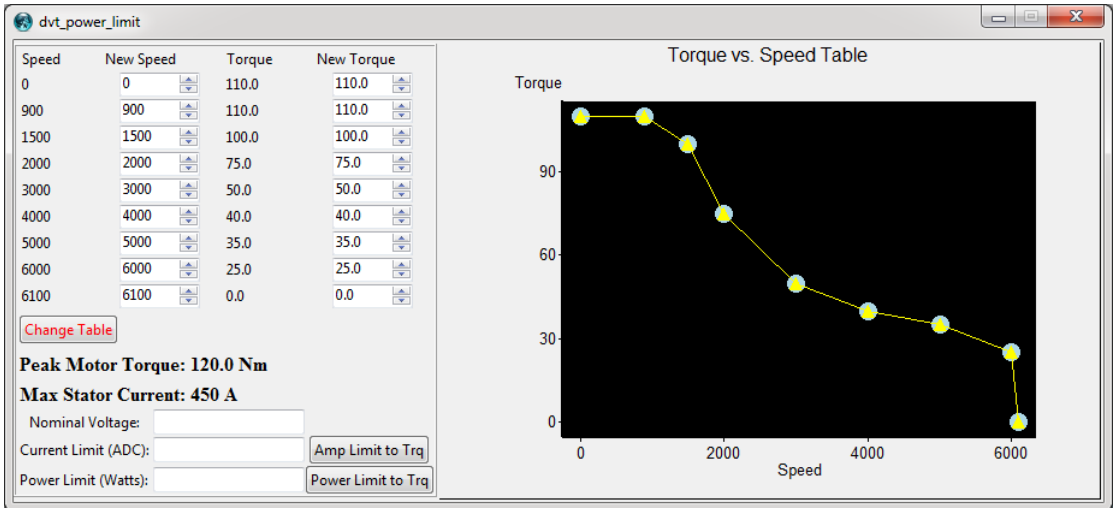


Figure 3 Motor power limit map in DVT Helper

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5. Flag 3 - Thermals

Besides motor control torque cutbacks and power limit cutbacks, cutbacks may be applied for protection from over-temperature. The torque will be reduced if either the motor or controller is too hot.

NOTE: this will **not** automatically reduce the fluxing current (induction) or the field weakening current (PMAC). For flux vector the reduction in torque request should demand less fluxing current due to the motor flux map and for PMAC the reduced torque should cause the motor to slow and reduce the field weakening current requirement. For the induction V/f algorithm the controller will attempt to output the motor rated magnetising current whenever there is a demand – even if the thermal cutback has reduced the torque limit to zero.

However, it would be good practice to adhere to these steps:

- For induction motors, make sure that the fluxing current for zero torque is not so high as to result in substantial heating of the motor at zero torque demand.
- For PMAC motors, ensure that field weakening currents that would normally be experienced at high speed are either not sustained for excessive periods of time, or will not result in excessive heating, during typical over-temperature conditions.

5.1. Motor over-temperature

If the motor gets too hot a “motor over-temperature” warning is set – provided the protection is enabled and configured correctly. For the motor the protection is user configurable through a map.



CAUTION there is **no over-temperature trip** for the motor. Setting a fluxing map (or minimum Id current) that is too high current for an induction motor, running a PMAC motor into field weakening continuously (where the currents are larger than the motor’s thermal capability) or setting the rated magnetising current too high when using the induction V/f algorithm will cause excessive heating and damage to the motor.

This 6 –point table is provided at object 0x461E, which allows the profile to be set. An example table, which cuts back over the range 110 to 120 deg C, would be as follows:

Temperature (deg C)	Cutback factor (p.u.)
110	1.0
112	0.8
114	0.6
116	0.4
118	0.2
120	0.0

The following limitations must be observed; otherwise a dynamic range error will occur:

- Temperature values must increase progressively through the table
- Cutback factors must decrease progressively through the table
- The first cutback factor must by 1.0
- The last cutback factor must be 0.0

For more information on setting up Genus motor temperature sensors see this app note.

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5.2.Controller over-temperature

As well as the motor over-temperature, the controller can get too hot. The controller cutback occurs over a 10C range and the temperatures are fixed. This therefore means that the controller temperature cutbacks are not configurable unlike the motor cutbacks.

If the controller does get too hot a “device too hot” warning is set. The first bits in the warning indicate the device causing the cutback. This is also found in the H-protocol Tdevice in message 12B00 and is also TPDO mappable in 0x5101, sub-index 45.

See 5.2.1 for HVLP and 5.2.2 Dragon8 over-temperature protection.

5.2.1. HVLP

Value	Device name
0x00	IGBT switching junction temperature estimation 1
0x01	IGBT switching junction temperature estimation 2
0x02	IGBT switching junction temperature estimation 3
0x03	IGBT switching junction temperature estimation 4
0x04	IGBT switching junction temperature estimation 5
0x05	IGBT switching junction temperature estimation 6
0x06	IGBT diode junction temperature estimation 1
0x07	IGBT diode junction temperature estimation 2
0x08	IGBT diode junction temperature estimation 3
0x09	IGBT diode junction temperature estimation 4
0x0a	IGBT diode junction temperature estimation 5
0x0b	IGBT diode junction temperature estimation 6
0x0c	Track temperature estimate 1
0x0d	Track temperature estimate 2
0x0e	Track temperature estimate 3
0x0f	Track temperature estimate 4
0x10	Track temperature estimate 5
0x11	Capacitor temperature estimation
0x12	Capacitor long term temperature estimation
0x13	Heatsink measured temperature
0x14	MCP measured track temperature
0x15	MCP CPU die temperature
0x16	MCP measured track temperature
0x17	IOP CPU die temperature
0x18	Capacitor measured temperature

5.2.2. Dragon 8

Value	Device name
0x00	MOSFET junction temperature estimation 1
0x01	MOSFET junction temperature estimation 2
0x02	MOSFET junction temperature estimation 3

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0x03	MOSFET junction temperature estimation 4
0x04	MOSFET junction temperature estimation 5
0x05	MOSFET junction temperature estimation 6
0x06	MOSFET junction delta temperature estimation 1
0x07	MOSFET junction delta temperature estimation 2
0x08	MOSFET junction delta temperature estimation 3
0x09	MOSFET junction delta temperature estimation 4
0x0a	MOSFET junction delta temperature estimation 5
0x0b	MOSFET junction delta temperature estimation 6
0x0c	Battery switch temperature estimate
0x0d	Track temperature estimate 1
0x0e	Track temperature estimate 2
0x0f	Track temperature estimate 3
0x10	Track temperature estimate 4
0x11	Track temperature estimate 5
0x12	Track temperature estimate 6
0x13	Capacitor temperature estimation
0x14	Capacitor long term temperature estimation
0x15	Heatsink measured temperature 1
0x16	Heatsink measured temperature 2
0x17	Heatsink measured temperature 3
0x18	MCP measured track temperature
0x19	MCP CPU die temperature
0x1a	IOP CPU die temperature

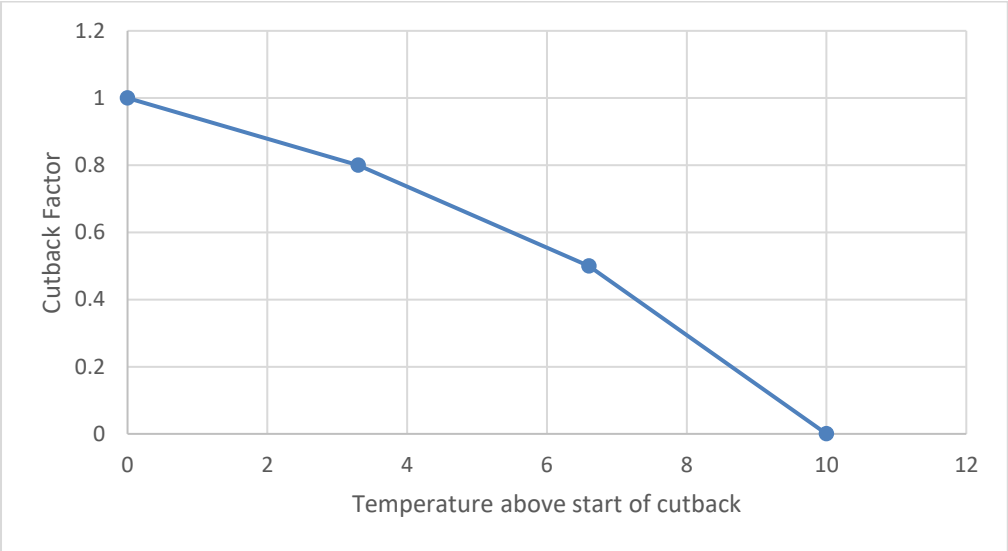
The controller hardware will trip and set a “Device Over-temperature” fault if the temperature continues to increase above full cutback. Again the first bits in the fault message indicate which hardware component is too hot.

Torque is also limited below -30C on the heatsink and a “device too cold” warning is set. All torque is disabled at -40C.

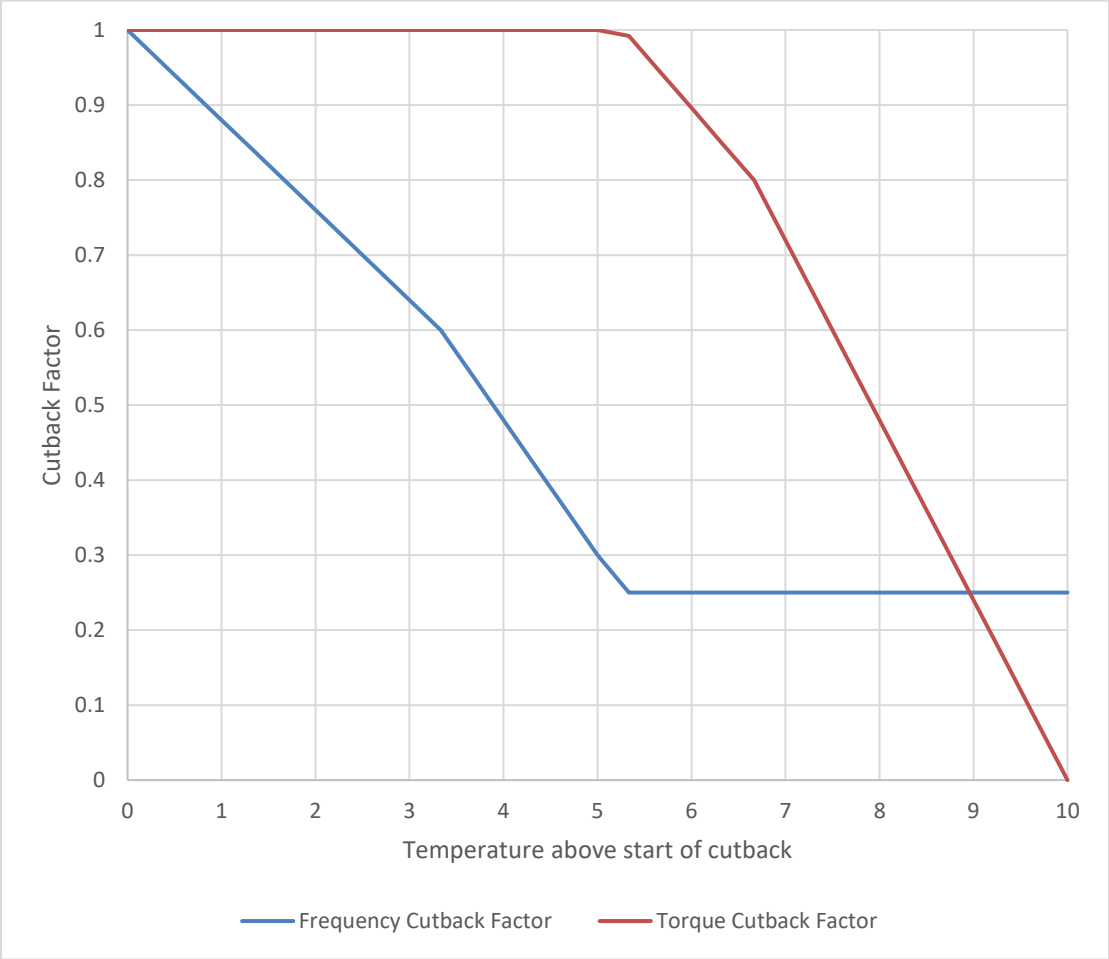
5.3. Reducing controller heating

It is possible to reduce the switching losses in the controller by adjusting the switching frequency and the switching waveform. Typically the cutback is applied as follows on the torque limits:

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It is possible to setup the controller to reduce switching frequency first before reducing the torque. This will only be applied to junction and heatsink cutbacks and only if the motor electrical frequency is 10 times lower than the switching frequency, otherwise the motor electrical frequency (x10) acts as an increased minimum switching frequency. For example with a maximum frequency of 8kHz and a minimum of 2kHz:



There is an additional option to reduce switching losses using a feature called Bus Clamping Modulation. This has to be either permanently enabled or disabled and is configured in the PWM flags 0x5831, sub-index 5, bit 2 (set the bit to enable). It may cause an increase in switching noise, depending upon the motor type.

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6. Flag 4 - User limits via H-protocol

The H protocol monitor in DVT is useful to check the sent over CAN or from TracApp / PumpApp. Below in Figure 4 Torque limits shown in the H-Protocol Monitor, the useful information about torque limits are highlighted. T_demand shows the current Torque Demand, any control - words that you have sent and torque and speed limits from TracApp. So for example, if the limits are set very low, consider changing these values in the driveability profiles or traction baseline profile.

J1939 Status Information

71-FA-11000 - T_demand	unknown	Nm	FF-71-12800 - Iq	38.125	A
71-FA-11000 - Controlword	unknown		FF-71-12800 - Id_ref	49.0625	A
71-FA-11000 - T_drive_limit	unknown	Nm	FF-71-12800 - Iq_ref	36.25	A
71-FA-11100 - T_regen_limit	unknown	Nm	FF-71-12900 - Ud	-0.125	V
71-FA-11100 - Fwd_speed_limit	unknown	RPM	FF-71-12900 - Uq	11.375	V
71-FA-11100 - Rev_speed_limit	unknown	RPM	FF-71-12900 - Mod_index	73.046875	%
71-FA-11200 - I_batt_discharge_limit	unknown	A	FF-71-12900 - T_control	20.3125	Nm
71-FA-11200 - I_batt_recharge_limit	unknown	A	FF-71-12A00 - Tj_0	26	DegC
71-FA-11200 - V_cap_target	unknown	V	FF-71-12A00 - Tj_1	26	DegC
71-71-11000 - T_demand	20.4375	Nm	FF-71-12A00 - Tj_2	26	DegC
71-71-11000 - Controlword	0		FF-71-12A00 - Tj_3	26	DegC
71-71-11000 - T_drive_limit	2047.9375	Nm	FF-71-12A00 - Tj_4	26	DegC
71-71-11100 - T_regen_limit	-2048.0	Nm	FF-71-12A00 - Tj_5	26	DegC
71-71-11100 - Fwd_speed_limit	5000	RPM	FF-71-12A00 - T_motor_meas	63	DegC
71-71-11100 - Rev_speed_limit	10	RPM	FF-71-12A00 - T_motor_est	0	DegC
71-71-11200 - I_batt_discharge_limit	32767	A	FF-71-12B00 - T_cap_fast	23	DegC
71-71-11200 - I_batt_recharge_limit	-32768	A	FF-71-12B00 - T_track	25	DegC
71-71-11200 - V_cap_target	0.0	V	FF-71-12B00 - T_cap_slow	21	DegC
FF-71-11800 - T_measured	7.5625	Nm	FF-71-12B00 - T_heatsink	24	DegC
FF-71-11800 - Motor_speed	1504	rpm	FF-71-12B00 - Tj_delta	2	DegC
FF-71-11800 - I_batt	39	A	FF-71-12B00 - T_device	255	
FF-71-11900 - T_max_forward	7.5625	Nm	FF-71-12B00 - T_coolant	0	DegC
FF-71-11900 - T_max_reverse	-99.75	Nm	FF-71-12B00 - T_winding	0	DegC
FF-71-11900 - Statusword	8		FF-71-12C00 - Max_flux	0	A
FF-71-11900 - Limit_code	7		FF-71-12C00 - Iv_lim	0	A
FF-71-11A00 - T_controller_remaining	63	degC	FF-71-12C00 - Iq_max	0	A
FF-71-11A00 - T_motor_remaining	63	degC	FF-71-12C00 - Ls	0.0	H
FF-71-11A00 - V_cap	36.875	V	71-71-14000 - OP1	unknown	application specific units
FF-71-11B00 - Fault_code	17862		71-71-14000 - OP2	unknown	application specific units
FF-71-11B00 - Fault_data	0		71-71-14000 - OP3	unknown	application specific units
FF-71-11C00 - Encoder_position	-451.93028259277344	rotations	71-71-14000 - OP4	unknown	application specific units
FF-71-12800 - Id	48.875	A	71-71-14100 - OP5	4095	application specific units

Start/stop logging | Stopped

Figure 4 Torque limits shown in the H-Protocol Monitor

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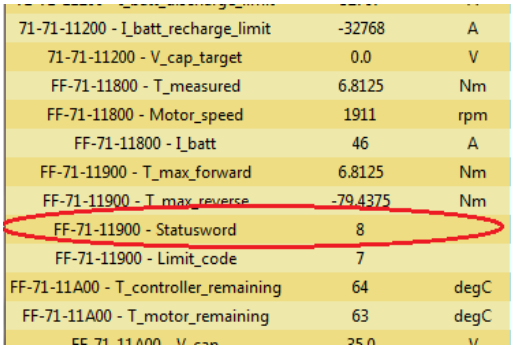
7. Flag 5 - 3D Lookup Table Limits

If configured as PMAC 3D look up table control, a torque limit map is also applied for each table and these tables can impact how well the motor drives. Consider updating the tables to reduce limits and loading back into the genus controller. Note: Genus lookup tables are now loaded using .DLD files - See app note.

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8. Flag 6 -Motor Control FSM State

If the motor control FSM state is not in the motor operational states (ENABLED) then all torque limits are forced to zero. You can see this in the H-Protocol monitor under status-word.



71-71-11200 - I_batt_recharge_limit	-32768	A
71-71-11200 - V_cap_target	0.0	V
FF-71-11800 - T_measured	6.8125	Nm
FF-71-11800 - Motor_speed	1911	rpm
FF-71-11800 - I_batt	46	A
FF-71-11900 - T_max_forward	6.8125	Nm
FF-71-11900 - T_max_reverse	-79.4375	Nm
FF-71-11900 - Statusword	8	
FF-71-11900 - Limit_code	7	
FF-71-11A00 - T_controller_remaining	64	degC
FF-71-11A00 - T_motor_remaining	63	degC

Figure 5 Motor Control FSM - Status word in H Protocol monitor

The status word could be either of the following:

- NOT READY TO SWITCH ON = 0x0001,
- SHUTDOWN = 0x0002,
- PRECHARGE = 0x0004,
- ENERGISED = 0x0007,
- ENABLED = 0x0008,
- FAULT_OFF = 0x000D.

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9. Flag 7 - DC link voltage

Unlike most previous products and firmware, the voltage cutback has been split into separate drive and regenerative braking maps. The cutback acts on the torque limits. The cutback is calculated every 1ms and allows the torque to be instantly reduced, but takes 2 seconds to recover once the voltage has returned to an acceptable value.

9.1.1. Drive map

It is important to note that the drive map usually only reduces torque at low DC link voltages. The Drive map can be viewed in the DVT helper.

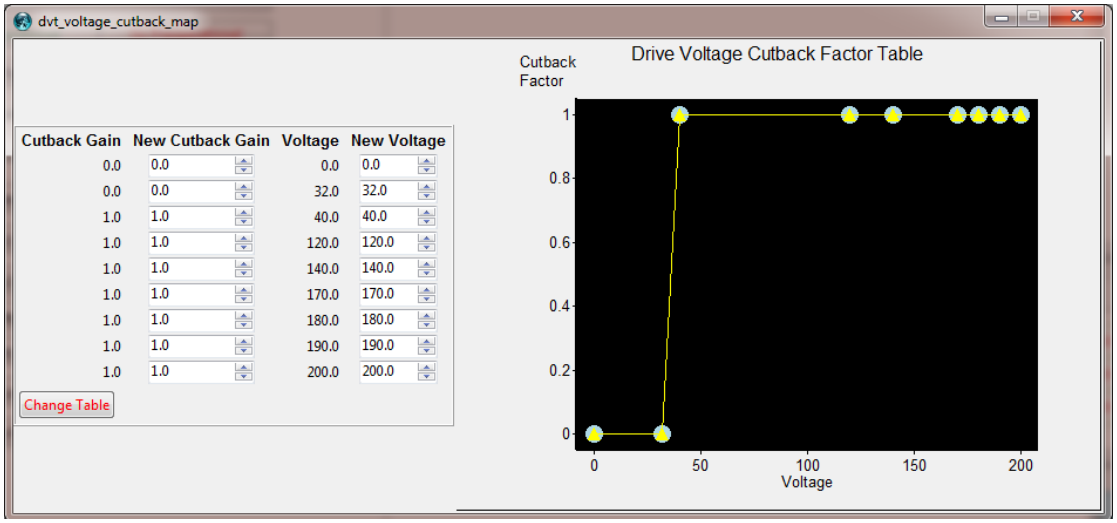


Figure 6 Drive Voltage Cutback Factor Table

9.1.2. Regen map

Unlike the drive map, the regen map only reduces torque at high DC link voltages. The regen map can be viewed and modified in the DVT helper.

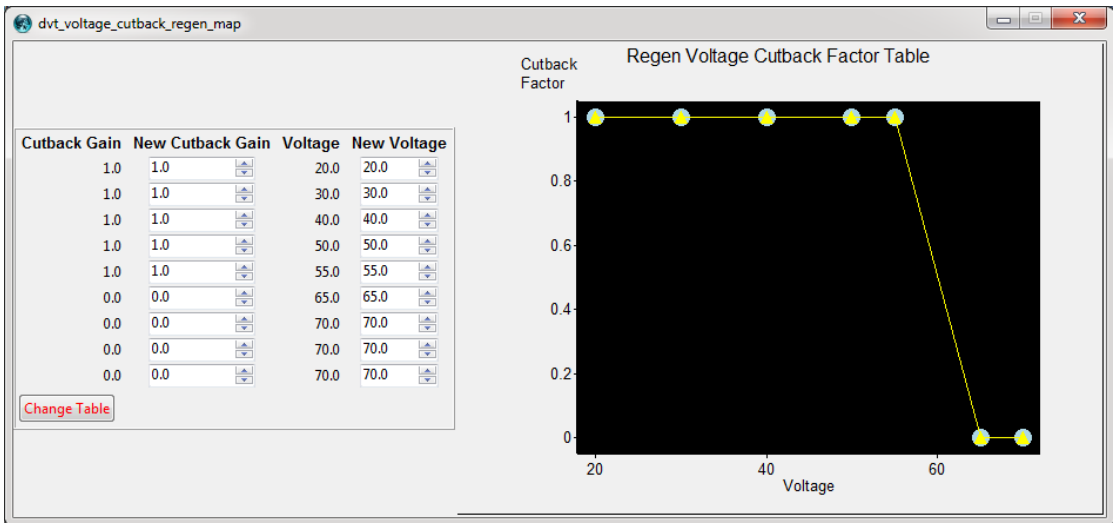


Figure 7 Regen voltage cutback factor table

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9.1.3. Voltage feedback filtering

For the voltage cutback mas above the raw DC link voltage measurement is used by default. It is possible to use a filtered voltage feedback with a fixed (non-configurable) 1ms filter time constant. This is useful if the DC link is especially noisy or subject to spikes in voltage, it is enabled by setting miscellaneous configuration parameters 2, 0x4650, sub-index 2, bit 12.

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10. Flag 8 - DC link current

If flag 8 is set then the cutback is being applied because of battery current. It is possible to limit the amount of battery / DC link current in 0x4623 in the DVT helper config tree.

You can set:

Item	Description
Maximum battery change current	Maximum allowed battery charge current for this motor
Maximum battery discharge current	Maximum allowed battery discharge current for this motor
Cutback range	Specifies where battery current cutback begins
Battery current ramp up rate	Ramp rate for cutback on battery current limit
Battery current ramp down rate	Ramp rate for cutback on battery current limit
Integral gain	Integral gain for battery current limit cutback
Minimum battery charge current speed	Speed below which discharge current limit is used instead of charge current limit

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11. Flag 9 - Speed limit

If the torque cutback flag 0x09 “speed limit” has been set, it is worth checking that the speed limits are sensible values. For more information on setting the speed limit parameters please refer to the following app note.

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